United International University
QUEST FOR EXCELLENCE

CSE 1326: Digital Logic Design Lab
Sequence recognizers

United International University

Objective

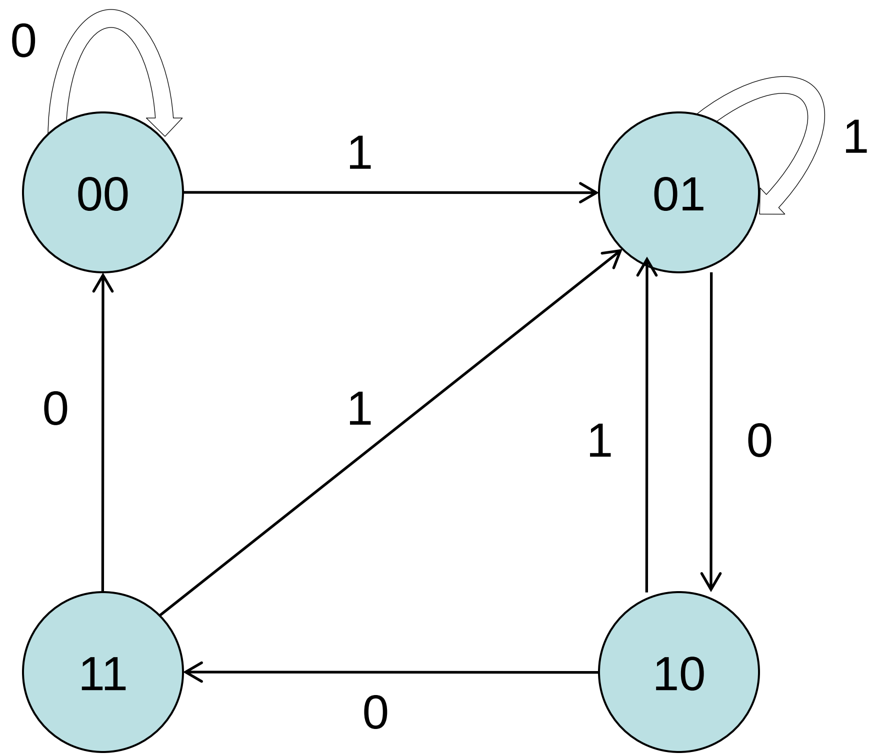
- Step 1: Implementing Sequence Recognizer using
  - **J-K Flip flops**
  - D Flip flops
  - Our example will detect the bit pattern 1001, while overlap will be supported

Inputs: 1 1 1 0 0 1 1 0 1 0 0 1 0 0 1 1 0 ...

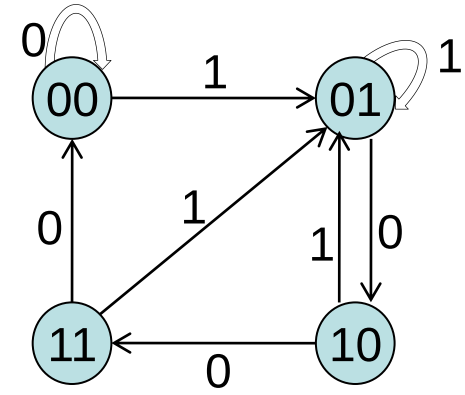
Outputs: 0 0 0 0 0 1 0 0 0 0 0 1 0 0 1 0 0 ...

  - If the pattern is detected/recognized, it will provide 1 to the output, otherwise 0.

Step 2: State Diagram



Step 3: State table



Present state		Input	Next state		Output	J1	K1	J0	K0
Q1	**Q0**	**X**	**Q1**	**Q0**	**Z**				
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	X
0	1	0	1	0	0	1	X	X	1
0	1	1	0	1	0	0	X	X	0
1	0	0	1	1	0	X	0	1	X
1	0	1	0	1	0	X	1	1	X
1	1	0	0	0	0	X	1	X	1
1	1	1	0	1	1	X	1	X	0

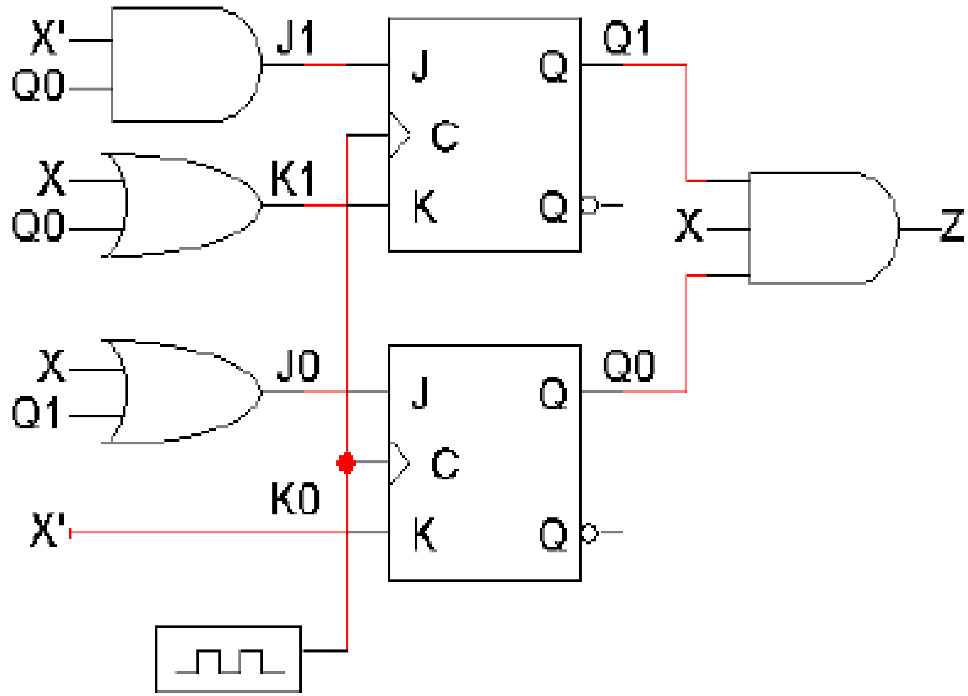
Implementing with J-K

- Equations and logic diagram

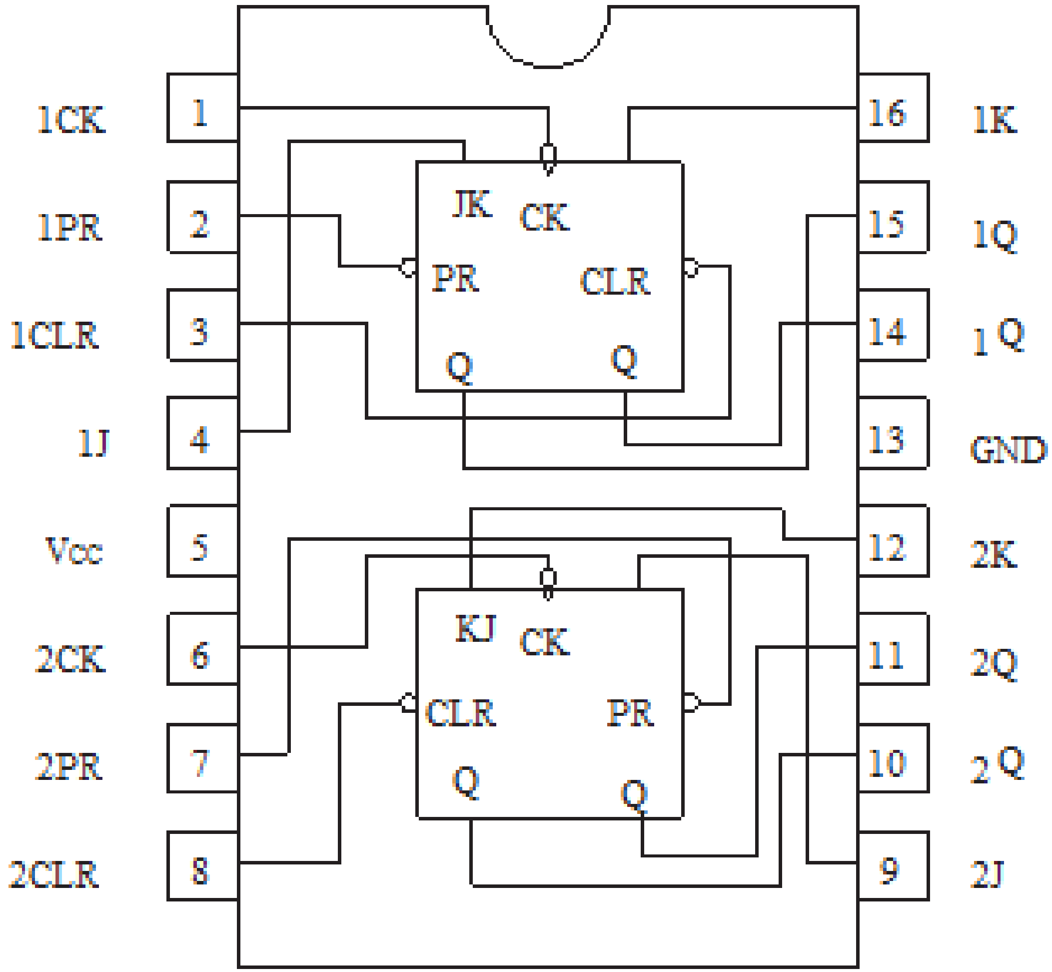
$J_1 = X' Q_0$
$K_1 = X + Q_0$

$J_0 = X + Q_1$
$K_0 = X'$

$Z = Q_1 Q_0 X$

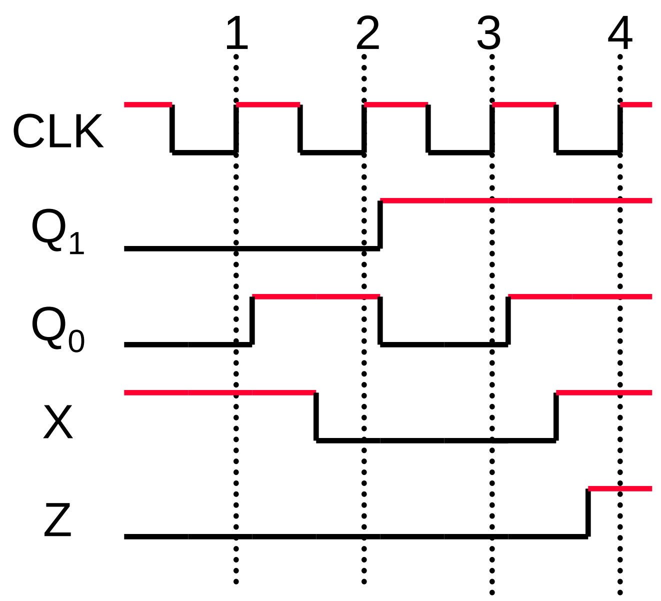


7476: Dual J-K Flip-Flops (with Preset and Clear)



Timing diagram

- Here is one example timing diagram for our sequence detector.
  - The flip-flops Q_1Q_0 start in the initial state, 00.
  - On the first three positive clock edges, X is 1, 0, and 0. These inputs cause Q_1Q_0 to change, so after the third edge $Q_1Q_0 = 11$.
  - Then when X=1, Z becomes 1 also, meaning that 1001 was found.
- The output Z does not have to change at positive clock edges. Instead, it may change whenever X changes, since $Z = Q_1Q_0X$.



Writing Report

- State diagram
- For both J-K and D
  - Used ICs with Pin diagram
  - State table with Flip-flop input assignment table
  - K-maps of each Functions
  - Circuit diagram